

Password Cracker

Questions and answers, Layer8 Orientation

Vault Lock 01

- Find the 4-digit vault code.
- All four digits add up to 26.
- The second digit is 4.
- The first digit is 5 more than the second digit.
- The third digit is 1 less than the fourth digit.

Answer: **9467**

Hint: The second digit is given. The first digit is 5 more than it.

Vault Lock 02

- Find the 4-digit vault code.
- The first digit is 4.
- The second digit is 2 more than the first digit.
- The second digit is 4 more than the third digit.
- The last two digits add up to 10.

Answer: **4628**

Hint: The first digit is given. Use it to get the second, then the third.

Vault Lock 03

Start with 18, then:

1. Multiply by 4
2. Add 36
3. Divide by 3
4. Multiply by 7
5. Subtract 84
6. Divide by 2
7. Add 16
8. Multiply by 12
9. Subtract 100

Answer: **1100**

Hint: After the first three steps you should have 36.

Vault Lock 04

Find the missing number in each sequence. The four missing numbers form the vault code. If a missing number has two digits, use only its last digit.

- 1st digit: 3, 6, 9, 12, ?
- 2nd digit: 2, 5, 10, 17, ?
- 3rd digit: 20, 17, 13, 8, ?
- 4th digit: 1, 2, 4, 7, ?

Answer: **5621**

Hint: 1st: add 3 each time. 2nd: add 3, 5, 7, 9. 3rd: take away 3, 4, 5, 6. 4th: add 1, 2, 3, 4.

Vault Lock 05

Find the 4-digit vault code. No digit is repeated.

- 1867: one digit is correct and in the correct position.
- 3109: two digits are correct and in the correct positions.
- 1732: no digit is correct.
- 5726: no digit is correct.

Answer: **4809**

Hint: 1732 and 5726 rule out 1, 7, 3, 2, 5 and 6. See what is left in 1867 and 3109.

Vault Lock 06

Find the 4-digit vault code. No digit is repeated.

- 2980: one digit is correct and in the correct position.
- 8402: no digit is correct.
- 1752: two digits are correct and in the correct positions.
- 4350: one digit is correct and in the correct position.

Answer: **1956**

Hint: 8402 rules out 8, 4, 0 and 2, so in 2980 the 9 must be the correct digit (second place).

Vault Lock 07

Start with 15, then:

1. Multiply by 4
2. Add 20
3. Divide by 2
4. Multiply by 6
5. Subtract 40
6. Multiply by 3
7. Add 25
8. Multiply by 2

Answer: **1250**

Hint: After four steps you should have 240.

Vault Lock 08

Start with 18, then:

1. Multiply by 5
2. Add 10
3. Divide by 2
4. Multiply by 8
5. Subtract 40
6. Multiply by 3
7. Add 20
8. Multiply by 2

Answer: **2200**

Hint: After four steps you should have 400.

Vault Lock 09

- Find the 4-digit vault code.
- All four digits add up to 19.
- The second digit is 2.
- The first digit is 4 more than the second digit.
- The third digit is 5 more than the second digit.
- The fourth digit is 2 less than the first digit.

Answer: **6274**

Hint: The first digit is 4 more than 2. Everything else follows from the first two digits.

Vault Lock 10

Find the missing number in each sequence. The four missing numbers form the vault code.

- 1st digit: 1, 2, 2, 4, ?
- 2nd digit: 20, 18, 14, 8, ?
- 3rd digit: 0, 1, 1, 2, 3, ?
- 4th digit: 25, 16, 20, 11, 15, ?

Answer: **8056**

Hint: 1st: multiply the previous two. 2nd: the gaps grow by 2. 3rd: add the previous two. 4th: two sequences take turns, each going down by 5.

Vault Lock 11

Find the 4-digit vault code. No digit is repeated.

- 8125: one digit is correct and in the correct position.
- 6907: two digits are correct and in the correct positions.
- 1732: no digit is correct.
- 5726: no digit is correct.

Answer: **8904**

Hint: 1732 and 5726 rule out 1, 7, 3, 2, 5 and 6. In 8125 only the 8 can be right.

Vault Lock 12

Find the 4-digit vault code. No digit is repeated.

- 4972: one digit is correct and in the correct position.
- 5230: two digits are correct and in the correct positions.
- 1289: no digit is correct.
- 8394: no digit is correct.

Answer: **5670**

Hint: 1289 and 8394 rule out 1, 2, 8, 9, 3 and 4. In 4972 only the 7 can be right.

Vault Lock 13

Start with 12, then:

1. Multiply by 6
2. Add 8
3. Divide by 4
4. Add 7
5. Multiply by 7
6. Subtract 9
7. Multiply by 6
8. Subtract 1

Answer: **1079**

Hint: After five steps you should have 189.

Vault Lock 14

- Find the 4-digit password.
- The first two digits multiply to 32.
- The last two digits multiply to 35.
- All four digits add up to 24.
- The first digit is smaller than the second digit.
- The third digit is greater than the fourth digit.

Answer: **4875**

Hint: Which two digits multiply to 35? Which two multiply to 32 with the smaller one first?

Vault Lock 15

Start with 100, then:

1. Subtract 25
2. Divide by 5
3. Multiply by 8
4. A hexagon has 6 sides. Multiply 6×1000 .
5. Add the result to your current number.

Answer: **6120**

Hint: Before the hexagon step you should have 120.

Vault Lock 16

Find the 4-digit code.

- The first digit is the number of letters in THREE.
- The second digit is the number of sides of a square.
- The third digit is the number of vowels in LAYER8.
- The fourth digit is the number of months in a year divided by 2.

Answer: **5426**

Hint: LAYER8 has two vowels: A and E.

Vault Lock 17

- Find the 4-digit password.
- It is not 1234.
- It is not 4321.
- It contains no 0.
- The first digit is greater than 5.
- The second digit is less than 5.
- The third digit is even.
- The last digit is odd.
- The first and last digits add up to 10.
- The second digit is exactly 2 more than the third digit.
- The second and last digits add up to 7.

Answer: **7423**

Hint: Use 'first + last = 10' and 'second + last = 7' together with 'second = third + 2'.

Vault Lock 18

Start with 200, then:

1. Multiply by 6
2. Subtract 172
3. Divide by 2
4. Multiply by 3
5. Add 18

Answer: **1560**

Hint: After two steps you should have 1028.

Vault Lock 19

Find the 4-digit code.

- Digit 1: the smallest prime number.
- Digit 2: the square root of 49.
- Digit 3: the number of letters in HACK.
- Digit 4: the number of sides on a hexagon.

Answer: **2746**

Hint: The smallest prime number is 2, not 1.

Vault Lock 20

Find the 4-digit password.

- Digit 1: number of letters in ONE.
- Digit 2: number of letters in TWO.
- Digit 3: number of letters in THREE.
- Digit 4: number of letters in FOUR.

Answer: **3354**

Hint: Just count the letters: O-N-E is 3.

Vault Lock 21

Find the 4-digit password.

- Digit 1: how many different letters appear more than once in MISSISSIPPI?
- Digit 2: how many months have 31 days?
- Digit 3: how many letters are in HACKER?
- Digit 4: how many sides does a STOP sign have?

⚠ One of these clues may not have the obvious answer you first expect.

Answer: **3768**

Hint: MISSISSIPPI: count the letters that repeat (I, S, P), not how often. A STOP sign is an octagon.

Vault Lock 22

Start with 48, then:

1. Multiply by 20
2. Add 240
3. Divide by 4
4. Subtract 137
5. Multiply by 6
6. Add 21
7. Add 1

Answer: **1000**

Hint: After four steps you should have 163.

Vault Lock 23

- Find the 4-digit password.
- All four digits are different.
- The password contains no 0, 1, 5 or 8.
- The first two digits add up to 10.
- The first digit is greater than the second.
- The last two digits add up to 11.
- The first and last digits add up to 9.

Answer: **7392**

Hint: The first two digits add to 10 and can't use 1, 5 or 8. Try $7 + 3$ or $6 + 4$.

Vault Lock 24

- Find the 4-digit password.
- All four digits are different.
- The first two digits add up to 13.
- The last digit is even.
- The first digit is twice the last digit.
- The second digit is 4 more than the third digit.
- The first digit is greater than 5.

Answer: **8514**

Hint: The first digit is twice the last and greater than 5, so it must be 8.

Vault Lock 25

- Find the 4-digit password.
- Exactly two digits are even.
- No digit is repeated.
- The first and last digits add up to 10.
- The middle two digits multiply to 24.
- The digits increase from left to right.
- The first digit is a prime number.

Answer: **3467**

Hint: Which two digits multiply to 24? The digits go up from left to right.

Vault Lock 26

Start with 72, then:

1. Multiply by 15
2. Subtract 180
3. Divide by 3
4. Multiply by 8
5. Add 64

Answer: **2464**

Hint: After three steps you should have 300.

Vault Lock 27

Start with 1000, then:

1. Add 80
2. Divide by 6
3. Add 75
4. Multiply by 8
5. Subtract 100
6. Divide by 5
7. Multiply by 13

Answer: **5044**

Hint: After the 'Divide by 6' step you should have 180.

Vault Lock 28

I am not a number, but I can become one. Find the letters where:

- Q comes before W
- E comes before R

Use the keyboard keys directly above these letters.

Answer: **1234**

Hint: Look at the number keys directly above Q, W, E and R.

Vault Lock 29

Find the 4-digit password.

- Digit 1: double the number of letters in HACK.
- Digit 2: number of sides on a pentagon + 1.
- Digit 3: number of letters in SCAM.
- Digit 4: number of authentication factors represented by 2FA.

Answer: **8642**

Hint: HACK has 4 letters. 2FA means two factors.

Vault Lock 30

Start with 1000, then:

1. Add 80
2. Divide by 6
3. Add 75
4. Multiply by 8
5. Subtract 100
6. Divide by 4
7. Multiply by 5

Answer: **2425**

Hint: After four steps you should have 2040.

Vault Lock 31

Start with 750, then:

1. Add 150
2. Divide by 5
3. Add 64
4. Multiply by 7
5. Subtract 100
6. Divide by 4
7. Multiply by 6

Answer: **2412**

Hint: After four steps you should have 1708.

Vault Lock 32

Start with 800, then:

1. Add 200
2. Divide by 5
3. Add 100
4. Multiply by 4
5. Subtract 200
6. Divide by 2
7. Multiply by 5

Answer: **2500**

Hint: After four steps you should have 1200.

Vault Lock 33

Start with 900, then:

1. Add 100
2. Divide by 5
3. Add 100
4. Multiply by 4
5. Subtract 200
6. Divide by 2
7. Multiply by 3

Answer: **1500**

Hint: After four steps you should have 1200.

Vault Lock 34

Start with 4821, then:

1. Reverse the number
2. Add 1000
3. Subtract 321

Answer: **1963**

Hint: 4821 reversed is 1284.

Vault Lock 35

Starting code: 1638, then:

1. Reverse the number
2. Subtract 2000

Answer: **6361**

Hint: 1638 reversed is 8361.

Vault Lock 36

Start with 8642, then:

1. Add the first and last digits.
2. Add the second and third digits.
3. Find the difference between the two results.
4. Multiply the difference by 100.
5. Add 2000.

Answer: **2000**

Hint: Both sums come out to 10, so the difference is 0.

Vault Lock 37

Start with 450, then:

1. Add 50
2. Multiply by 5
3. Subtract 500
4. Multiply by 4

Answer: **8000**

Hint: After three steps you should have 2000.

Vault Lock 37 (Part 2)

- The password has four digits.
- The first digit is 2 more than the second digit.
- The third digit is twice the second digit.
- The fourth digit is 1 more than the first digit.
- The second digit is 3.

Answer: **5366**

Hint: Start from the second digit: it is 3.

Vault Lock 38

- The code is not 1111.
- The code is not 9999.
- The first digit is odd.
- The second digit is even.
- The third digit is less than the second.
- The last digit is less than the first.
- The sum of all digits is 20.
- No digit is repeated.
- The second digit is the largest digit.
- The third digit is 1 more than the last digit.
- The last digit is odd.

Answer: **5843**

Hint: The digits add to 20 and the second is the largest. Try 8 for the second digit.

Vault Lock 39

- 4927: one digit correct and in the right place, and two digits correct but in the wrong place.
- 8136: one digit correct but in the wrong place.
- 2751: three digits correct, two of them in the right place.
- 6408: nothing is correct.
- The first digit is even.

Answer: **2971**

Hint: 6408 rules out 6, 4, 0 and 8. 2751 shares three digits with the code.

Vault Lock 40

Start with 40, then:

1. Multiply by 5
2. Add 300
3. Divide by 10
4. Multiply by 9
5. Subtract 60
6. Multiply by 2
7. Add 220

Answer: **1000**

Hint: After three steps you should have 50.

Vault Lock 41

- Digit 1: letters in "CYBER".
- Digit 2: vowels in "SECURITY".
- Digit 3: number of digits in a PIN.
- Digit 4: factors of 10 (count).

Answer: **5344**

Hint: The factors of 10 are 1, 2, 5 and 10.

Vault Lock 42

- No digit repeats.
- The code contains no 1, 5, or 9.
- First digit = $2 \times$ third digit.
- Second digit = first digit $-$ 2.
- Third digit is even.
- Last digit is even.
- Sum of all digits = 18.

Answer: **8640**

Hint: The first digit is twice the third, and the third is even. Try 4 for the third digit.

Vault Lock 43

Start with 30, then:

1. Multiply by 6
2. Add 180
3. Divide by 3
4. Multiply by 5
5. Subtract 150
6. Divide by 2
7. Multiply by 6

Answer: **1350**

Hint: After three steps you should have 120.

Vault Lock 44

- ❌ 1234: nothing is correct.
- 🔥 5073: two digits correct, one in the right place.
- 🎯 5890: two digits correct and in the right place.
- ❌ 7012: one digit correct but in the wrong place.
- 🔥 9687: three digits correct, one in the right place.
- No repetition.

Answer: **5867**

Hint: 1234 rules out 1, 2, 3 and 4, so in 7012 only the 7 or the 0 can be right.

Vault Lock 45

- Digit 1: days in a week.
- Digit 2: sides in a triangle.
- Digit 3: letters in "CODE".
- Digit 4: legs on a spider.

Answer: **7348**

Hint: A spider has 8 legs.

Vault Lock 46

- All digits different.
- Sum = 19.
- First digit is 3 more than the second.
- Third digit is half of the fourth.
- Second digit is 5.

Answer: **8524**

Hint: The second digit is 5, so the first is 8.

Vault Lock 47

Start with 12, then:

1. Multiply by 9
2. Add 108
3. Divide by 6
4. Multiply by 7
5. Subtract 42
6. Multiply by 5

Answer: **1050**

Hint: After three steps you should have 36.

Vault Lock 48

- The code is not 2468.
- The code is not 1357.
- First digit is even.
- Second is odd.
- Third is less than the second.
- Last is greater than the first.
- Sum = 18.
- No digit is repeated.
- Second is the largest digit.
- Third is half the first.
- Last is odd.

Answer: **4725**

Hint: The third digit is half the first. The last is odd and bigger than the first.

Vault Lock 49

- Digit 1: number of bits in a nibble.
- Digit 2: number of layers in the OSI model.
- Digit 3: number of letters in "DATA".
- Digit 4: number of digits in an HTTP status code.

Answer: **4743**

Hint: A nibble is half a byte: 4 bits. The OSI model has 7 layers.

Vault Lock 50

Start with 50, then:

1. Multiply by 4
2. Add 200
3. Divide by 5
4. Multiply by 6
5. Subtract 120
6. Multiply by 3

Answer: **1080**

Hint: After three steps you should have 80.

Vault Lock 51

The password contains four different digits. You receive these attempts:

- 5837: exactly 2 digits are correct and in the correct position. No other digit appears in the password.
- 6429: exactly 1 digit is correct and in the correct position. No other digit appears in the password.
- 1578: exactly 3 digits are correct, but all three are in the wrong positions.
- 9045: exactly 1 digit is correct, but it is in the wrong position.
- 1027: exactly 1 digit is correct and in the correct position, and exactly 1 other digit is correct but in the wrong position.

Answer: **5821**

Hint: 5837 and 6429 tell you which digits are definitely out. Then use 1578 to place the rest.

Vault Lock 52

Start with 80, then:

1. Multiply by 5
2. Add 2600
3. Divide by 10
4. Multiply by 6
5. Subtract 120

Answer: **1680**

Hint: After two steps you should have 3000.

Vault Lock 53

- Digit 1: number of letters in "LOGIN".
- Digit 2: number of vowels in "PASSWORD".
- Digit 3: number of digits in an OTP.
- Digit 4: number of factors of 6.

Answer: **5264**

Hint: PASSWORD has two vowels: A and O. An OTP is usually 6 digits.

Vault Lock 54

Start with 36, then:

1. Multiply by 7
2. Add 84
3. Divide by 6
4. Multiply by 25
5. Subtract 60

Answer: **1340**

Hint: After three steps you should have 56.

Vault Lock 55

- All digits different.
- First digit = $2 \times$ second.
- Third digit = second $\div$ 2.
- Fourth digit = first $-$ 2.
- Sum = 20.

Answer: **8426**

Hint: Work backwards: the third digit is half the second, and the second is half the first.

Vault Lock 56

- No repetition.
- Contains no 0 or 9.
- First digit $>$ second.
- Third digit is odd.
- Last digit is prime.
- First + last = 10.
- Second + third = 8.
- Second $<$ third.
- Second digit is prime.

Answer: **8352**

Hint: First + last = 10 and the last is prime. Try 2 or 3 for the last digit.

Vault Lock 57

- Digit 1: number of letters in "PATCH".
- Digit 2: vowels in "FIREWALL".
- Digit 3: number of sides on a hexagon.
- Digit 4: prime numbers between 1 and 3 (count).

Answer: **5362**

Hint: FIREWALL has three vowels: I, E and A. The primes from 1 to 3 are 2 and 3.

Vault Lock 58

- Digit 1: the hex value for the letter 'A' is 41. Take the second digit.
- Digit 2: the decimal ASCII code for a newline character (`\n`) is 10. Take the last digit.
- Digit 3: the standard port number for SSH (22). Add the two digits together.
- Digit 4: the binary representation of decimal 3 (0011). Count the number of zeros.

Answer: **1042**

Hint: SSH uses port 22. Decimal 3 is 0011 in binary.

Vault Lock 59

- Digit 1: output of an AND gate when Input A = 1 and Input B = 1.
- Digit 2: output of an OR gate when Input A = 0 and Input B = 0.
- Digit 3: output of a NAND gate when Input A = 1 and Input B = 1.
- Digit 4: output of a NOT gate when Input A = 0.

Answer: **1001**

Hint: AND needs both inputs to be 1. NOT flips the input.

Vault Lock 60

Hot = right digit in the right place. Warm = right digit in the wrong place.

- 1357 → 1 Hot, 2 Warm
- 7218 → 1 Hot, 0 Warm
- 4960 → 0 Hot, 1 Warm
- 2386 → 1 Hot, 0 Warm
- 5370 → 1 Hot, 3 Warm

Answer: **7305**

Hint: 5370 is 1 hot + 3 warm, so the code uses exactly the digits 5, 3, 7 and 0.

Vault Lock 61

Hot = right digit in the right place. Warm = right digit in the wrong place. Cold = digit not in the code.

- 9124 → 1 Hot, 3 Warm
- 5830 → Everything is Cold
- 2678 → 1 Hot, 0 Warm
- 4192 → 0 Hot, 4 Warm

Answer: **2914**

Hint: 4192 is 4 warm, so the code uses exactly the digits 4, 1, 9 and 2.

Vault Lock 62

- Digit 1: a pizza has 8 slices. You eat 3 and your friend eats 2. How many slices are left?
- Digit 2: a cricket over has 6 balls. 4 balls have been bowled. How many are left in the over?
- Digit 3: you go to the gym every other day, starting Monday, for one week (Monday to Sunday). How many gym days is that?
- Digit 4: you buy a ₹91 snack with a ₹100 note. How many rupees of change do you get?

Answer: **3249**

Hint: Gym days: Monday, Wednesday, Friday, Sunday.

Vault Lock 63

Use an old phone keypad: 2=ABC, 3=DEF, 4=GHI, 5=JKL, 6=MNO, 7=PQRS, 8=TUV, 9=WXYZ. The code is the word "SAFE" typed on this keypad.

Answer: **7233**

Hint: S is on key 7 and A is on key 2.

Vault Lock 64

Start with 35, then:

1. Multiply by 6
2. Subtract 30
3. Divide by 4
4. Add 17
5. Multiply by 3
6. Multiply by 6

Answer: **1116**

Hint: After three steps you should have 45.

Vault Lock 65

Start with 45, then:

1. Multiply by 4
2. Subtract 60
3. Divide by 6
4. Multiply by 15
5. Add 70
6. Multiply by 3
7. Subtract 11

Answer: **1099**

Hint: After four steps you should have 300.

Vault Lock 66

Look at your keyboard's top row: Q=1, W=2, E=3, R=4, T=5, Y=6, U=7, I=8, O=9, P=0. The code is the word "WIRE" typed using those keys as numbers.

Answer: **2843**

Hint: W is the 2nd key on the top row and I is the 8th.

Vault Lock 67

- Digit 1: V
- Digit 2: IX
- Digit 3: III
- Digit 4: VII

Answer: **5937**

Hint: These are Roman numerals: V is 5.

Vault Lock 68

On a standard die, opposite faces always add up to 7. You can see these top faces: 1, 3, 5, 2. The code is the number on the bottom face of each die, in the same order.

Answer: **6425**

Hint: Take each top face away from 7.

Vault Lock 69

The code is the atomic numbers of these elements, in order: Helium, Carbon, Oxygen, Hydrogen.

Answer: **2681**

Hint: Helium is element 2 and Hydrogen is element 1.

Vault Lock 70

- Digit 1: at exactly 3:00, which number does the hour hand point at?
- Digit 2: at 4:25, which number does the minute hand point at?
- Digit 3: at exactly 6:00, which number does the hour hand point at?
- Digit 4: at 7:40, which number does the minute hand point at?

Answer: **3568**

Hint: At 25 minutes past, the minute hand points at the 5.

Vault Lock 71

Each group of 4 bits is one digit of the code: 0111 | 0011 | 1000 | 0100

Answer: **7384**

Hint: Convert each group from binary: 0111 is 7.

Vault Lock 72

Work out each line. The four results, in order, form the code:

- Digit 1: the number of days in a week minus the number of weekend days.
- Digit 2: the number of legs on a spider divided by the number of legs on a human.
- Digit 3: the number of sides on a hexagon plus the number of sides on a triangle.
- Digit 4: the number of wheels on a car minus the number of wheels on a unicycle.

Answer: **5493**

Hint: There are 2 weekend days. A unicycle has 1 wheel.

Vault Lock 73

- The password has four different digits.
- The first digit is twice the fourth digit.
- The second digit is 1 more than the third digit.
- The sum of all four digits is 18.
- The first digit is greater than 5.

Answer: **6543**

Hint: The first digit is greater than 5 and twice the last, so it is 6 or 8.

Vault Lock 74

Start with 25, then:

1. Multiply by 8
2. Subtract 50
3. Divide by 5
4. Add 12
5. Multiply by 3
6. Multiply by 10
7. Add 5

Answer: **1265**

Hint: After four steps you should have 42.

Vault Lock 75

- Find the 4-digit password.
- All four digits are different.
- All four digits add up to 23.
- The first digit is 7.
- The second digit is 5 less than the first digit.
- The third digit is odd.
- The fourth digit is 4 more than the third digit.

Answer: **7259**

Hint: The second digit is $7 - 5 = 2$.
